Soft Switch Lock-Release Mechanism for a Switch-Mode Hydraulic Pump Circuit

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Switch-mode hydraulic circuits are a theoretically efficient, compact, fast responding, and inexpensive control option. Despite the many potential benefits of switch-mode hydraulic circuits, the control method suffers from large energy losses during transitions of the high-speed valve due to throttling and fluid compressibility. Rannow and Li previously proposed utilizing soft switching to minimize the throttling energy loss (Rannow and Li, "Soft Switching Approach to Reducing Transition Losses in On/Off Hydraulic Valve," J. Dyn. Syst., Measure. Control (in press)). A major challenge of this approach is a locking soft switch that releases quickly and with precise timing, while under load. In this paper, a novel soft switch locking mechanism is presented that utilizes the pressure signal in the switched volume to trigger the release. A dynamic model is developed of three unique soft switch circuits and two control circuits that create a virtually variable displacement pump. The model is used to perform a grid search optimization of the soft switch parameters for the three circuits. The three soft switch circuits reduce the throttling and compressibility energy losses between 49% and 66% compared with the control circuit. The simulation results demonstrated that the soft switch circuits perform as expected for duty cycles and pressures below the design conditions. At higher duty cycles and pressures, the short time the circuit is connected to tank prevented the soft switches from resetting between cycles, preventing proper function. This novel lock and release soft switch mechanism enables soft switching in switch-mode hydraulic circuits, which significantly reduces throttling and compressibility energy losses during valve transitions. Lower losses during valve transition allow the use of slower switching valves, lowering energy consumption, and cost. [DOI: 10.1115/1.4026327]

Keywords: switch-mode hydraulics, digital hydraulics, soft switching, hydraulic control

1 Introduction/Background

Switch-mode hydraulic circuits, the hydraulic analog of switch-mode control from power electronics [1], have been proposed as an alternative to conventional hydraulic control methods such as metering valves and variable displacement pumps. Digital hydraulic circuits are theoretically more efficient than metering valve control and less expensive and bulky than variable displacement pump control [2]. In switch-mode control, a high-speed valve is rapidly switched between completely on and off states. The power reaching the load is determined by the duty cycle of the valve, defined as the fraction of the switching period the valve is in the on position. Switch-mode control allows any pump, motor, or hydraulic actuator to have a virtually variable displacement.

Despite the excellent potential of switch-mode hydraulics, the technology has not penetrated the marketplace, partially due to the low realized efficiency. Generally speaking, there are five major sources of energy loss in the valve portion of a switch-mode hydraulic circuit: internal leakage, viscous friction, throttling during fully-open and transitioning events, inertial forces in oscillating valve designs, and fluid compressibility; note that the losses in the pump, motor, or actuator are typically neglected for generality as the additional losses are unit specific. Work at Purdue University and by Tomlinson et al. modeled the steady and transitional throttling losses and the compressibility loss, using the assumption of a constant bulk modulus [3–5]. Work by Li's research group has considered all of the above mentioned energy loss separately, utilizing a pressure dependent compressibility term [6]. The author further expanded on these works by modeling the fluid compressibility and throttling loss together to capture their interaction [7]. In these various works, the fluid throttling loss has been found to create 15–60% of the total energy losses, depending on the circuit and valve design [6,8].

The throttling energy loss in a switch-mode hydraulic circuit is dominated by the valve transition events. Consider a virtually variable displacement pump (VVDP) circuit using a 3-way high-speed valve, shown in Fig. 1(*a*), and the equivalent circuit using two 2-way high-speed valves, shown in Fig. 1(*b*). While a VVDP circuit can be created with a single 2-way valve, the 3-way valve allows flow reversal for operation as a hydraulic motor. During the valve transitions, the valve ports decrease in area to zero, before opening to the other position. This causes the pressure in the switched volume to rise and the flow from the fixed displacement pump to be throttled across the partially open valve, generating throttling energy losses. The pressure rise in the switched volume during valve transitions can be managed by placing a relief valve between the switched volume and tank or by placing a check valve between the switched volume and pressure rail, both of which are shown in Fig. 1(*b*). Rannow and Li recently proposed a novel solution to minimize the transition throttling energy losses, by absorbing the flow that would otherwise pass through the valve during the transition in a soft switch [9].

Conceptually, the soft switch, shown in Fig. 1(*c*), consists of two small chambers with spring-loaded pistons. The piston in one of the chambers contains an externally actuated lock, which is released at a specific time in the switching cycle. At the start of a switching cycle, when the tank valve is fully open, both soft switch chambers are empty. When the tank valve is closing, the pressure in the switched volume begins to increase and overcomes the force of the spring in the passive soft switch chamber, causing

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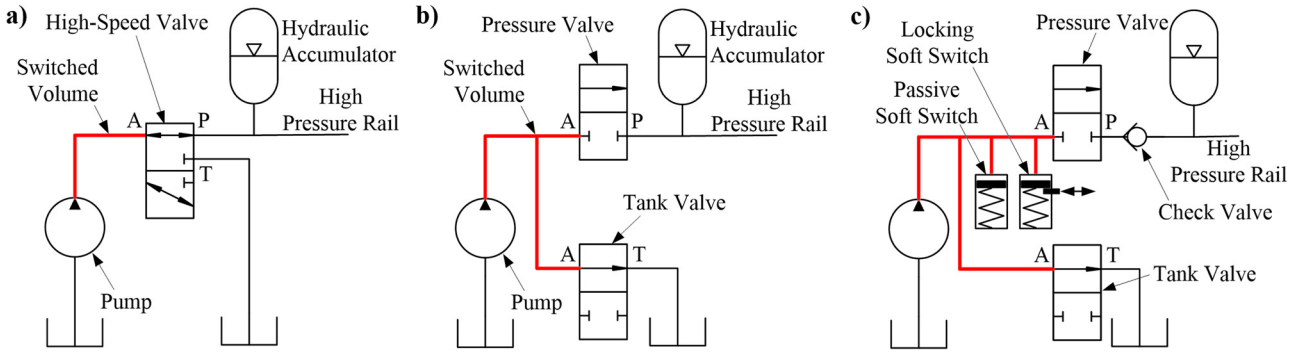

Fig. 1 Circuit (a) is a generic virtually variable displacement pump circuit. Circuit (b) is equivalent to circuit (a), but uses two 2-way valves instead of a single 3-way valve. The relief valve and check valve illustrate two ways of reducing pressure rises during valve transitions. Circuit (c) illustrates the soft switching concept.

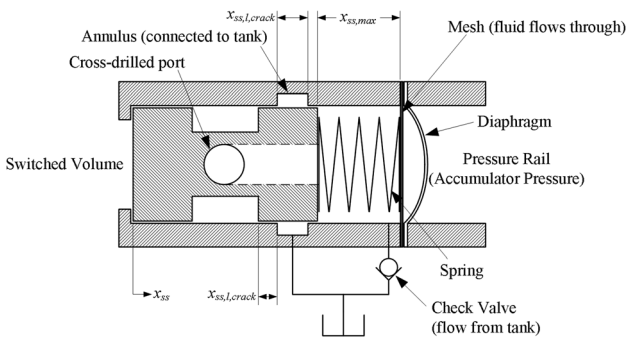

Fig. 2 Cross-sectional view of the novel soft switch lock and release mechanism

it to absorb flow. As the pressure valve opens, the soft switch continues to fill and reaches the maximum volume once the valve is fully open. During this time, the check valve prevents reverse flow through the pressure valve. As the pressure valve closes during the next transition, the locking soft switch is released and it begins to absorb flow. As the tank valve opens, the locking soft switch continues to fill, reaching the maximum volume when the tank valve is fully open. Once the tank valve is fully open, both soft switches empty to tank due to the force of the springs on the pistons. The two soft switch functions could be combined into a single soft switch with a locking mechanism at midstroke, as originally proposed by Rannow and Li [9]. In this work, two separated soft switches will be utilized, one free and the other locking, to simplify the implementation. A second difference is that Rannow and Li proposed a check valve in parallel with the pressure valve, instead of in series. In the design proposed in this paper, the soft switches absorb all of the flow during the valve transitions, while in the prior work, flow is only absorbed during transitions of the tank valve. This change is necessary for the actuation of the locking mechanism described below.

The barrier that has prevented physical implementation of the soft switch concept is the challenging requirements of the lock release mechanism. As noted by Rannow and Li, the soft switch lock must rapidly release with precise timing while under load [9]. In a typical digital hydraulic circuit, the valve transition time occupies 0.001 to 0.02 s, and thus the soft switch lock release must operate in a fraction of this time. A possible approach, proposed by Rannow and Li, is to use an electrically controlled active lock release mechanism [9]. To eliminate the need for an electromechanical lock with extremely fast response and precise timing, the novel lock solution presented in this paper uses the pressure signal in the switched volume for actuation.

The focus of this paper is on presenting a novel locking mechanism to enable soft switching in switch-mode hydraulic circuits. In Sec. 2, the locking mechanism is presented and three unique pump circuits are discussed. A dynamic model of the three soft switch circuits and two control circuits is developed in Sec. 3, followed by a presentation of simulation results in Sec. 4. The results are discussed in Sec. 5, followed by concluding remarks in Sec. 6.

2 Soft Switch Lock and Release Mechanism

To meet the soft switch lock and release requirements of fast actuation, precise timing, low actuation energy, and releasing under load, a novel hydraulic pressure based solution is proposed. The soft-switch locking mechanism, shown in Fig. 2, utilizes the balance of pressure forces on the soft switch piston to release the piston and absorb flow at the correct time in the cycle. As seen in the circuit diagram in Fig. 3(*a*), the left side of the piston is connected to the switched volume while the right side of the piston is connected to a pilot line from the high pressure rail. When the pressure valve begins to close, the pressure in the switched volume increases. This pressure increase causes a force imbalance on the piston that moves it to the right, deforming the diaphragm connected to the high-pressure rail. Once the piston has traveled a sufficient distance, a port to tank is opened, allowing fluid to flow from the right side of the piston through an interior passage in the piston to tank. The opening of this port drops the pressure on the right side of the piston (the diaphragm moves into contact with the rigid mesh), effectively releasing the hydraulic lock. This allows the soft switch to absorb flow and prevent throttling fluid across the partially open high-speed valves. Once the pressure valve has closed and the tank valve is fully open, the spring on the right side of the piston pushes the soft switch back to the lock position, sending the absorbed flow to tank. A check valve between tank and the right cavity of the piston allows the soft switch piston to reach the fully locked position. Once locked, the soft switch is ready for the next cycle.

The lock and release mechanism, illustrated in Fig. 2 and in a circuit in Fig. 3(*a*), is effective at releasing at the appropriate time in the switching cycle to absorb flow during the high-speed valve transitions, yet does have a significant shortcoming. During the initial travel of the locking soft switch piston, pressurized fluid is stored in the diaphragm. As the soft switch piston displacement opens the port to tank, this stored fluid is throttled across this orifice, resulting in a loss of this stored energy. This issue can be addressed by implementing a concept analogous to the check valve in parallel to the pressure valve illustrated in Fig. 1(*b*). Instead of using a diaphragm on the back side of the locking soft switch to allow communication with the accumulator pressure, this volume can be connected to the high pressure rail with a check valve as illustrated in Fig. 3(*b*). By using a check valve to the pressure rail instead of the diaphragm, the volume of fluid

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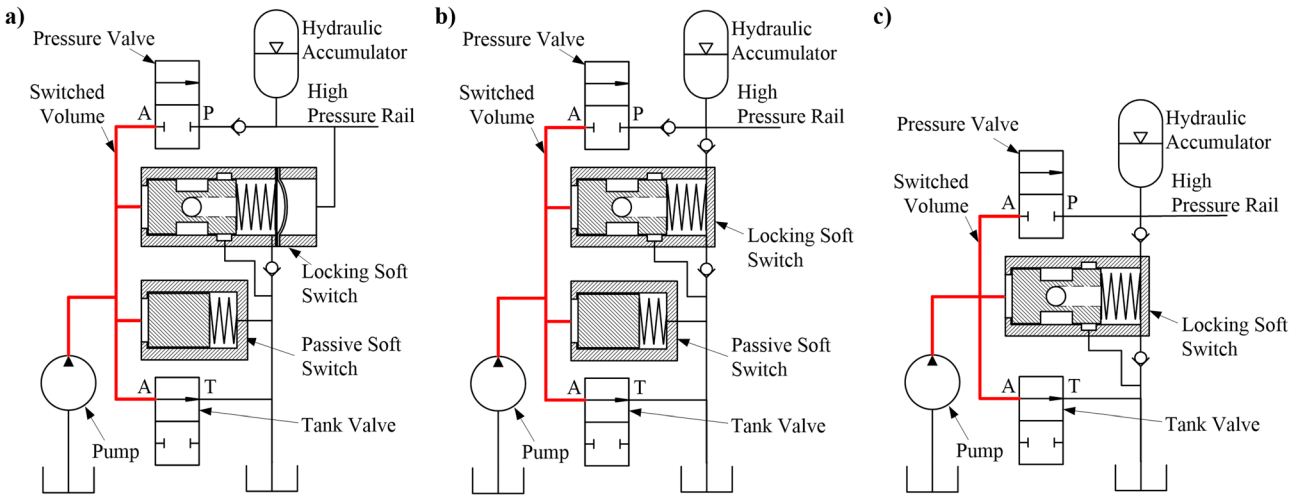

Fig. 3 Diagrams of the virtually variable displacement pump soft switch circuits a–c

displaced by the piston before the port opens to tank is pumped to the load and not stored to be throttled across the port to tank.

In consideration of the circuit in Fig. 3(*b*), further simplifications can be made at the expense of minor additional energy loss. By eliminating the diaphragm and using a check valve between the back side of the locking soft switch piston and the pressure rail, the displacement of the soft switch piston does not result in an energy loss. Thus, the distance the piston travels before opening the port to tank, $x_{\text{crack},sb}$, can be increased such that the pressure valve is fully closed by the time the soft switch port to tank is open. This longer travel allows elimination of the check valve in series with the high pressure valve, which makes transitions from pumping mode to motoring mode easier. Elimination of this check valve does not impact the transition event where the pressure valve is closing and the tank valve is opening. However, for the transition event where the tank valve is closing and the pressure valve is opening, elimination of this check valve results in reverse flow from the pressure rail through the pressure valve as it is opening. To minimize the loss associated with this flow, the pressure should rise as quickly as possible in the switched volume, dictating eliminating the passive soft switch, as shown in Fig. 3(*c*).

While the focus of this paper is on the pumping circuit, the concept of utilizing both passive and active locking soft switches can also be applied to switch-mode motoring circuit, Fig. 4. The function of the soft switches in the virtually variable displacement motor circuit are slightly different than the pump circuit in that the role of the soft switches is to supply flow to the switched volume instead of absorbing flow to minimize the throttling energy loss. Starting with the switched volume connected to the high pressure rail, both soft switches are to the far right positions and filled. As the pressure valve closes and tank valve opens, the passive soft switch supplies flow to the switched volume at a pressure slightly lower than the accumulator pressure. The check valve in series with the tank valve prevents flow reversal. Once the passive soft switch is empty, flow to the motor passes through the fully open tank valve. As the tank valve begins to close, the pressure in the switched volume drops, resulting in displacement of the locking soft switch piston. Once the piston displacement reaches the cracking position, the right side of the piston is exposed to the accumulator pressure, resulting in a rapid increase in the pressure of the switched volume. The locking soft switch continues to provide flow until the pressure valve is fully open. When the pressure valve is fully open, both soft switches refill and are ready for the next cycle.

As in the pumping circuit, the motoring circuit can either utilize check valves or a diaphragm on the back side of the locking soft switch. In addition, if a check valves are used to the back of the soft switch, the check valve in series with the tank valve can be eliminated to make the transition from pumping to motoring easier. An alternative method of enabling transitions between pump and motor modes is to use pilot-to-open check valves in series with the pressure and tank valves. The pilot signal can be used to defeat one of the two check valves for mode switching.

3 Modeling

A dynamic model is now presented that will be used to study the influence of the various soft switch architectures on behavior of a switch-mode circuits. The model is developed for five unique virtually variable displacement pump circuits: the relief valve circuit, the check valve in parallel with the high pressure valve circuit, both illustrated in Fig. 1(*b*), and the three soft switch architectures. The equations developed below are for the soft switch circuits, while comments will be provided about the construction of the other two circuit models. The methods presented are also applicable to other switch-mode circuits, such as a virtually variable displacement motor. The model captures the pressure dynamics in the switched volume and back side of the locking soft switch as well as the piston dynamics of the two soft switch

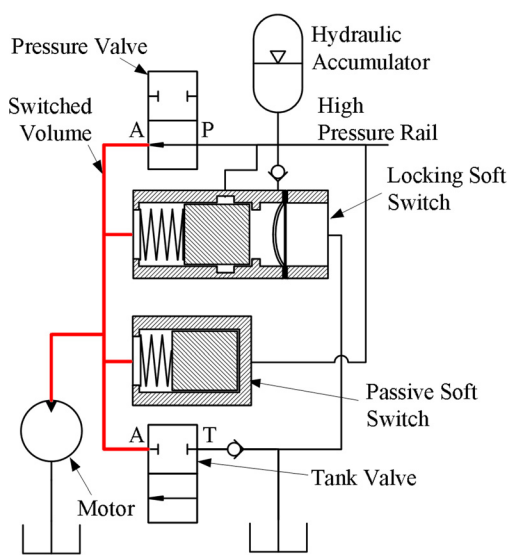

Fig. 4 Illustrating the implementation of the pressure-release soft switch concept on a virtually variable displacement motor circuit

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pistons. Note that while the flow passages in the switched volume appear long in the diagrams, this is only to clarify the circuit. In a physical implementation, these flow passages would be extremely short to minimize the volume. Based on this, the influence of fluid inertia and the resistance of the flow passages in the switched volume are assumed negligible. The pump is assumed to be a constant flow source and the high-pressure rail is assumed to be at a constant pressure due to the presence of the accumulator.

From the definition of the fluid bulk modulus, $\beta_e = -V(dP/dV)$, the pressure in the switched volume can be expressed as

$$\dot{P}_{\text{switch}} = \frac{\beta_e(P_{\text{switch}})}{(V_{\text{switch}} + A_{ss,p}x_{ss,p} + A_{ss,l}x_{ss,l})} \times (Q_{\text{pump}} - Q_{\text{valve},A\to P} - Q_{\text{valve},A\to T} - A_{ss,p}v_{ss,p} - A_{ss,l}v_{ss,l}) \tag{1}$$

where V_{switch} is the switched volume, $A_{ss,x}$ is the area of soft switch piston, $x_{ss,x}$ is the displacement of the soft switch piston, Q_{pump} is the flow rate of the pump, $Q_{\text{valve},A\to P}$ is the flow rate through the pressure valve, $Q_{\text{valve},A\to T}$ is the flow rate through the tank valve, and $v_{ss,x}$ is the velocity of the soft switch piston. The subscript x is replaced with p and l for the passive and locking soft switches respectively. In the relief valve and check valve circuits there are additional terms for the flow through these valves and the soft switch terms are eliminated. Note that the leakage flow through the valves and soft switches is assumed to be negligible. The effective bulk modulus, β_e, is modeled using the pressure-dependent equation developed by Cho et al. [10]

$$\beta_e(P) = \beta\left[\frac{\left(\frac{P}{P_o}\right)^{\frac{1}{\gamma}} e^{\frac{(P_o-P)}{\beta}} + R}{\frac{R}{\gamma}\frac{\beta}{P} + \left(\frac{P}{P_o}\right)^{\frac{1}{\gamma}} e^{\frac{(P_o-P)}{\beta}}}\right] \tag{2}$$

where β is the bulk modulus of air free oil, P_o is the atmospheric pressure, γ is the ratio of specific heats for air, and R is the volume fraction of air entrained in the oil.

The flow rate through the high-speed valve to tank is modeled using the orifice equation with a time varying area

$$Q_{\text{valve},A\to T} = C_D A_{A\to T}\sqrt{2/\rho|P_{\text{switch}} - P_{\text{tank}}|}\,\text{sgn}(P_{\text{switch}} - P_{\text{tank}}) \tag{3}$$

where C_D is the discharge coefficient, $A_{A\to T}$ is the A→T port valve area, plotted as a function of time in Fig. 5, ρ is the mass density of oil, and P_{tank} is the tank pressure. Note that this equation is derived from Bernoulli's equation, carrying the assumption of steady flow. While this is not strictly true in this case, the focus of this work is on comparing energy losses between circuits and thus the assumption is acceptable. The pressure valve is modeled in a similar way with the exception that it is in series with a check valve and thus their resistance can be combined as

$$\begin{aligned} &\text{if } P_{\text{switch}} - P_{\text{high}} > P_{\text{crack}} \\ &\quad \times Q_{\text{valve},A\to P} = \frac{C_D^2 A_{A\to P} A_{\text{check}}}{\sqrt{C_D^2 A_{A\to P}^2 + C_D^2 A_{\text{check}}^2}} \\ &\quad \times \sqrt{2/\rho(P_{\text{switch}} - P_{\text{high}} - P_{\text{crack}})} \\ &\text{else} \quad Q_{\text{valve},A\to P} = 0 \end{aligned} \tag{4}$$

where P_{high} is the accumulator pressure, which is assumed constant, P_{crack} is the check valve cracking pressure, and $A_{\text{check},P}$ is the area of the check valve. The check valve is assumed to open instantaneously when the pressure drop across the valve exceeds the cracking pressure. For the relief valve circuit, check valve circuit, and soft switch circuit c, Eq. (4) is modified by removing the conditional, including the absolute value under the radical and signum function to allow flow reversal, and eliminating the resistance of the check valve. In the relief valve and parallel check valve circuits, an additional flow rate term is added to Eq. (1) and the flow rate through the relief valve or check valve are modeled with a pressure based conditional.

The high-speed valves are assumed to have linearly moving spools with circular orifices, generating the open areas shown in Fig. 5 and further described in Ref. [7]. Note that the valve areas are representative of a commercially available valve. The four transition events are labeled t_1–t_4 and will be used throughout the paper.

The velocity and position of the soft switch pistons are found by applying Newton's second law to the pistons

$$\dot{v}_{ss,x} = \frac{(P_{\text{switch}} - P_{\text{back},x})A_{ss,x} - F_{\text{preload},x} - k_x x_{ss,x} - \frac{\mu v_{ss,x} 2\pi r_{ss,x} l_{ss,x}}{c_{ss,x}}}{m_{ss,x}} \tag{5}$$

$$\dot{x}_{ss,x} = v_{ss,x} \tag{6}$$

where $x_{ss,x}$ and $v_{ss,x}$ are the position and velocity of the piston, respectively, $F_{\text{preload},x}$ and k_x are the is the preload and rate of the spring respectively, μ is the dynamic viscosity of the fluid, $r_{ss,x}$ and $l_{ss,x}$ are the radius and length of the piston, $c_{ss,x}$ is the radial clearance of the piston in the bore, and $m_{ss,x}$ is the piston mass. The soft switch piston displacements are limited to a range from 0 to $x_{ss,x,\max}$ through a conditional statement.

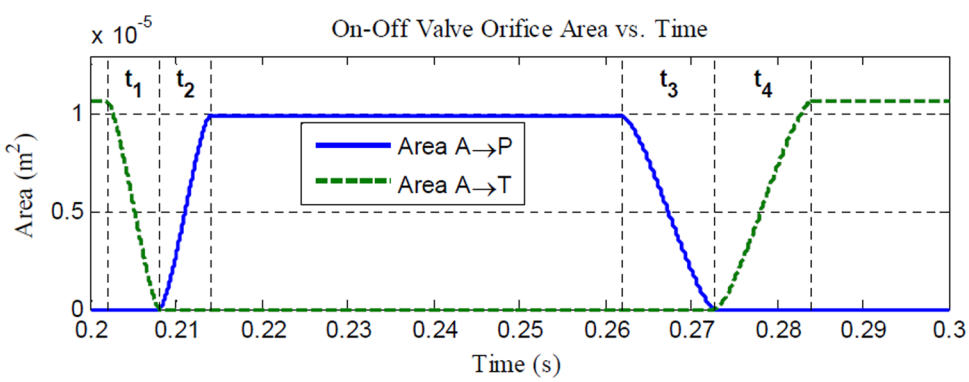

Fig. 5 High-speed valve orifice areas as a function of time during a switching period with a duty cycle of 60%. The vertical gridlines correspond with the beginning and end of each phase of the valve transition.

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As the back side of the passive soft switch piston is connected to a case drain to manage leakage, the pressure on the back side, $P_{\text{back},p}$, is equal to the tank pressure. The pressure on the back side of the locking soft switch piston is more complex. In soft switch circuit a, the back side of the soft switch piston communicates with the high-pressure rail through a diaphragm and the orifice to tank with an area that is a function of the piston travel. The back side pressure is determined by evaluating if the diaphragm is engaged, as explained in pseudo-code as

$$\begin{aligned}&\text{if } V_{\text{diaph}} > 0 \quad P_{\text{back},l} = P_{\text{high}}\\&\text{else} \quad \dot{P}_{\text{back},l} = \frac{\beta_e\left(P_{\text{back},b}\right)}{V_{ss,l,\text{clear}} + A_{ss,l}\left(x_{ss,l,\max} - x_{ss,l}\right)}\\&\quad \times \left(A_{ss,l}\dot{x}_{ss,l} - Q_{ss,l,T} + Q_{ss,l,\text{check},T}\right)\end{aligned} \tag{7}$$

where V_{diaph} is the volume of fluid stored in the diaphragm, $V_{ss,l,\text{clear}}$ is the clearance volume on the back side of the piston when it is at the maximum travel, $Q_{ss,l,T}$ is the flow rate out of the volume through the port to tank, and $Q_{ss,l,\text{check},T}$ is the flow rate from the tank to the back side of the piston through the check valve. Again, the pressure dependent effective bulk modulus for the fluid on the back side of the soft switch piston is found from Eq. (2). In the case of soft switch circuits b and c, Eq. (7) is revised to eliminate the diaphragm dependent conditional and include the flow rate out of the back of the soft switch volume to the pressure rail, $Q_{ss,l,\text{check},P}$.

The flow rates into and out of the back side of the locking soft switch piston are found by treating the port to tank and check valves as orifices

$$Q_{ss,l,T} = C_D A_{ss,l,T}\sqrt{2/\rho\left|P_{\text{back},l} - P_{\text{tank}}\right|}\,\text{sgn}\left(P_{\text{back},l} - P_{\text{tank}}\right) \tag{8}$$

$$\begin{aligned}&\text{if } P_{\text{tank}} - P_{\text{back},l} > P_{\text{crack}}\\&\quad \times Q_{ss,l,\text{check},T} = C_D A_{\text{check}}\sqrt{2/\rho\left(P_{\text{tank}} - P_{\text{back},l} - P_{\text{crack}}\right)}\\&\text{else} \quad Q_{ss,l,\text{check},T} = 0\end{aligned} \tag{9}$$

$$\begin{aligned}&\text{if } P_{\text{back},l} - P_{\text{high}} > P_{\text{crack}}\\&\quad \times Q_{ss,l,\text{check},P} = C_D A_{\text{check}}\sqrt{2/\rho\left(P_{\text{back},l} - P_{\text{high}} - P_{\text{crack}}\right)}\\&\text{else} \quad Q_{ss,l,\text{check},P} = 0\end{aligned} \tag{10}$$

Note that the check valves are assumed to open instantaneously and that Eq. (10) only applies to soft switch circuits b and c. The area of the orifice to tank, with reference to Fig. 2, is described as

$$\begin{aligned}&\text{if } x_{ss,l} < x_{ss,l,\text{crack}} \quad A_{ss,l,T} = 0\\&\text{else if } x_{ss,l} < x_{ss,l,\text{crack}} + l_{ss,l,T} \quad A_{ss,l,T} = 2\pi r_{ss,l}\left(x_{ss,l} - x_{ss,l,\text{crack}}\right)\\&\text{else} \quad A_{ss,l,T} = 2\pi r_{ss,l} l_{ss,l,T}\end{aligned} \tag{11}$$

where $l_{ss,l,T}$ is the axial length of the port to tank. For soft switch circuit a, the volume absorbed by the diaphragm is calculated as

$$\text{if } P_{\text{back},l} \geq P_{\text{high}} \quad \dot{V}_{\text{diaph}} = A_{ss,l}\dot{x}_{ss,l} - Q_{ss,l,T} + Q_{ss,l,\text{check}} \tag{12}$$

To compare the five circuits, the energy into and out of the circuit is required. The energy from the pump into the switch-mode circuit during one second of operation is defined as

Table 1 Geometric and operating parameters for the virtually variable displacement pump circuits

Parameter	Symbol	Units	Value				
			Relief circuit	Check circuit	Soft switch circuit a	Soft switch circuit b	Soft switch circuit c
Switching frequency	f	Hz	10				
Pump flow rate	Q_{pump}	l/min	4				
Switched volume	V_{switched}	cm^3	15				
Entrained air fraction at P_o	R	fraction	0.02				
Accumulator pressure	P_{high}	MPa	20.7				
Atmospheric pressure	P_o	kPa	101				
Tank pressure	P_{tank}	kPa	205				
Mass density of oil	ρ	kg/m^3	876				
Dynamic viscosity of oil	η	Pa*s	0.0404				
Bulk modulus of air free oil	β	GPa	1.9				
Ratio of specific heats for air	γ	unitless	1.4				
High-speed valve parameters:							
A→P discharge coeff × area	$C_D \times A_{A\to P}$	mm^2	5.98				
A→T discharge coeff × area	$C_D \times A_{A\to T}$	mm^2	6.35				
Delay before A→P transition	$t_{d,P}$	ms	2				
Delay before A→T transition	$t_{d,T}$	ms	2				
Transition time to A→P	t_P	ms	12				
Transition time to A→T	t_T	ms	22				
Check/Relief valve parameters:							
Check valve cracking pressure	P_{crack}	kPa	NA	20.7			
Check valve $C_D \times$ Area	$C_D \times A_{\text{check}}$	mm^2	NA	7.375			
Relief valve cracking pressure	P_{relief}	MPa	24.1	NA	NA	NA	NA
Soft switch parameters:							
Piston radius[a]	$r_{ss,x}$	mm	NA	NA	4.5		
Max travel of passive piston[a]	$x_{ss,p,\max}$	mm	NA	NA	0.6	0.6	0
Max travel of locking piston[a]	$x_{ss,l,\max}$	mm	NA	NA	10.6	17.4	11
Spring preload force[a]	$F_{\text{preload},x}$	N	NA	NA	17.6	19	17.5
Travel to open tank port on locking piston[a]	$x_{ss,l,\text{crack}}$	mm	NA	NA	3.8	1	3.6
Clearance volume on back of locking piston	$V_{ss,l,\text{clear}}$	cm^3	NA	NA	1		
Radial clearance of piston	$c_{ss,x}$	μm	NA	NA	10		

[a]Optimized parameter.

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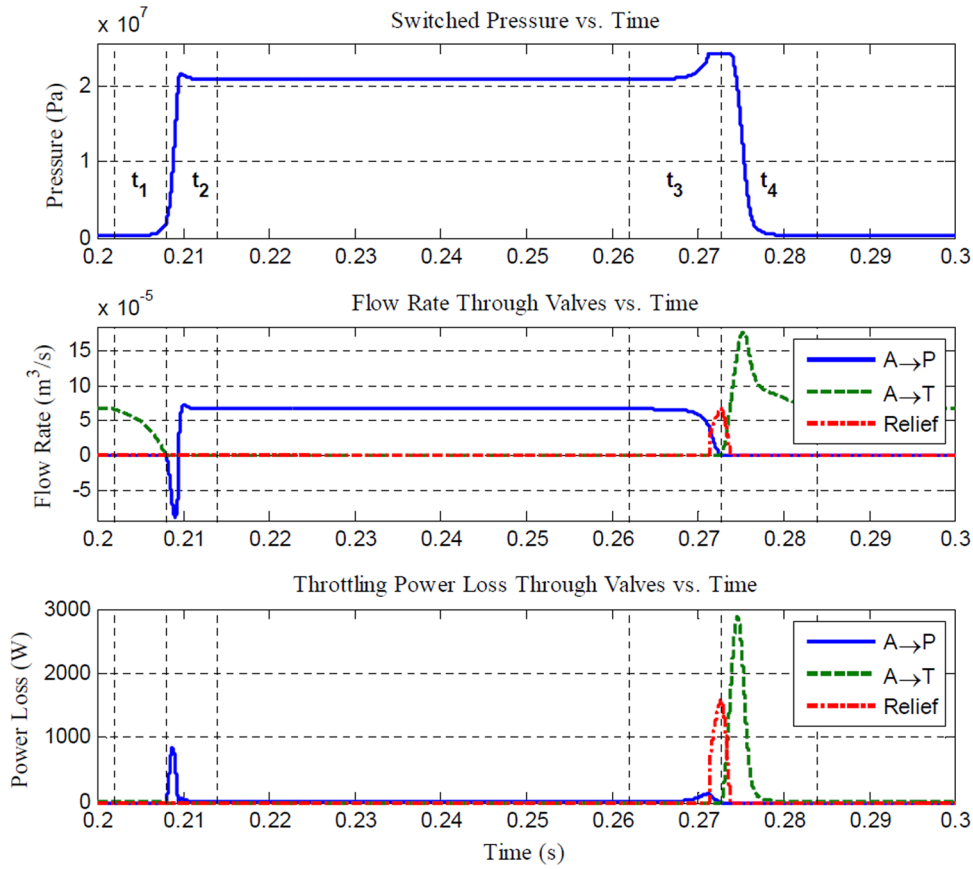

Fig. 6 Simulation results for the relief valve circuit illustrating the pressure in the switched volume, flow rate through the various valves, and throttling power loss as a function of time during a single switching cycle at a duty cycle of 60%. Note, the vertical gridlines mark the beginning and end of the valve transitions shown in Fig. 5.

$$E_{\text{pump}} = \int_0^1 (P_{\text{switch}} - P_{\text{tank}}) Q_{\text{pump}} dt \tag{13}$$

Similarly, the energy reaching the load during one second of operation is defined as

$$E_{\text{out}} = \int_0^1 \left(P_{\text{high}} - P_{\text{tank}}\right)\left(Q_{\text{valve},A\to P} + Q_{ss,l,\text{check},P}\right) dt \tag{14}$$

As the energy output from the system at a specific duty cycle does change with circuit architecture, the energy loss divided by the energy delivered by the pump, defined as the energy loss factor, is primarily used for comparison. Note that the efficiency is not reported as there are multiple other sources of energy loss in a switch-mode circuit that are not considered in this analysis, such as the leakage and valve actuation energy, and thus could be misleading in the context of comparison to other works. For this reason, the analysis will focus on the throttling and compressibility energy losses captured in the above calculations.

4 Results

The model developed in Sec. 3 will now be used to compare the five circuits and illustrate the dynamic behavior of the soft switching circuits. As the computation time for the simulation is quite short, a grid search method was used. In this method, the model was run for all combinations of operating parameters soft switch parameters in a gridlike fashion. The parameters varied in the optimization include a common radius for both chambers, the maximum displacement of each piston, the preload of both springs, and the distance the locking soft switch piston travels before opening the port to tank. The objective function was minimizing the energy loss per second of operation, found by subtracting Eq. (13) from Eq. (14). Geometric constraints were placed on the problem, defined by the values given in Table 1.

The optimal parameters for a given pressure and flow rate vary with the duty cycle. At low duty cycles, the system is connected to tank for a large portion of the switching period, providing a long time to empty the soft switch chambers. Conversely, at high duty cycles, the time for the soft switch chambers to discharge decreases. For the parameter optimization, a valve command duty cycle of 75% was chosen, ensuring that at any lower duty cycle the soft switch chambers fully empty during each cycle.

The simulations will use parameters for a generic virtually variable displacement pump circuit. The circuit uses a solenoid valve switching at 10 Hz, a fixed displacement pump with a flow rate of 4 l/min, a pressure rail set to 20.7 MPa, a switched volume of 15 cm^3, and a tank pressurized to 205 kPa. The length of the soft switch pistons is set equal to the diameter of the piston, to provide a sufficient bearing ratio. The piston mass is estimated by considering the density of steel and the piston volume found from the radius and length. To simplify the number of optimization variables, constant force springs are used and the spring rate is set to zero. Finally, the check valves used in the various circuits are all set to the same cracking pressure and fully open area. The fixed and optimized circuit parameters are presented in Table 1.

The simulation results of the control circuit, which utilizes a relief valve between the switched volume and tank, at a duty cycle of 60% and valve timing shown in Fig. 5, are presented in Fig. 6. During transition t_1, the port to tank closes, reducing the flow to tank and causing the pressure in the switched volume to rise, as

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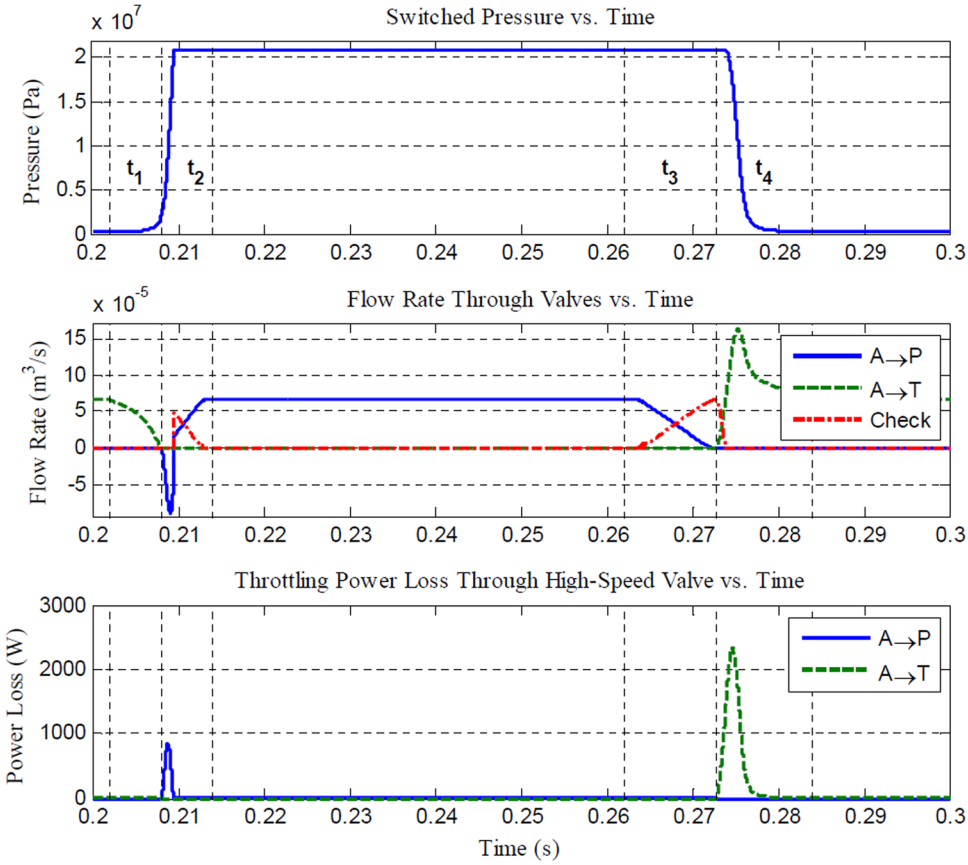

Fig. 7 Results of the simulation for the circuit with a check valve in parallel with the pressure valve at a duty cycle of 60%

determined by the compressibility of the fluid. As the port to pressure begins to open, transition t_2, the pressure in the switched volume has not yet reached the accumulator pressure, resulting in throttling of fluid from the accumulator into the switched volume and an associated energy loss. The function of the relief valve can be seen in transition t_3, where the closing of the pressure valve results in the switched volume pressure reaching the relief valve setting of 24.1 MPa. Note that the relief valve would also activate during transitions t_1 and t_2 if these transition event were longer or the pump flow rate was higher. As the tank valve opens, event t_4, the pressure in the switched volume slowly drops to near tank pressure. The throttling energy loss to the tank port during this transition dominates all of the throttling energy loss events. At the 60% duty cycle presented in the plot, the hydraulic energy produced by the pump during second is 926.2 J and the hydraulic energy reaching the high pressure rail is 828.9 J during one second, for an energy loss fraction of 0.105. Recall that the energy loss fraction is defined as the difference of the pump energy to the delivered energy divided by the pump energy.

The throttling energy loss associated with the flow across the relief valve can largely be eliminated by replacing the relief valve with a check valve in parallel with the pressure valve. As seen in Fig. 7, the check valve minimizes the pressure spikes in the switched volume during transition by passing flow to the high pressure rail. Note however, that while the energy loss associated with throttling across the relief valve is eliminated, significant throttling still occurs across the pressure valve due to the reverse flow as the pressure valve is opening and across the tank valve during opening to tank. Minimizing these two energy losses is the focus of soft switching. At 60% duty cycle, the hydraulic energy produced by the pump per second is 914.5 J and the hydraulic energy reaching the load per second is 857.8 J, for an energy loss fraction of 0.062.

The throttling energy losses across the pressure and tank valves observed in the relief and check valve circuit can be significantly reduced using soft switching. Simulation results of soft switch circuit a with the diaphragm, Fig. 3(*a*), are presented in Fig. 8. At the start of the cycle, the tank valve is fully open and both soft switches are empty. When the tank valve is closing, event t_1, the passive soft switch fills, minimizing throttling to the partially open tank valve. During event t_2, the pressure in the switched volume is increasing as the fluid is compressed and the pressure valve opens. Note that no flow passes through the high-speed valve until the switched volume pressure exceeds the pressure rail pressure due to the check valve in series with the pressure valve. During this period, the fluid behind the locking soft switch piston is also being compressed, as illustrated by the slight displacement of this piston. Both of these effects cause the slope of the pressure rise to be lower than the previous two cases. When the pressure valve closes, event t_3, the pressure in the switched volume rises and the locking soft switch piston is displaced. Once the piston reaches the cracking displacement, 1 mm, the pressure on the back side of the piston drops rapidly, allowing the soft switch to absorb fluid and avoid throttling fluid across the partially open pressure valve. The absorption of flow by the locking soft switch continues into event t_4, where the tank valve is opening. Both soft switches then empty and the cycle repeats. As seen in the fourth subplot, the throttling power loss across the tank and pressure valves is nearly two orders of magnitude lower than the relief and check valve cases. However, there is a large amplitude, short duration spike in the throttling power loss across the locking soft switch tank port that reaches 4500 W during event t_3 due to the discharge of the fluid stored in the diaphragm. For one second of operation of this circuit at 60% duty cycle, the energy supplied by the pump is 815.9 J and the energy to the high pressure rail is 778.6 J, for an energy loss fraction of 0.046.

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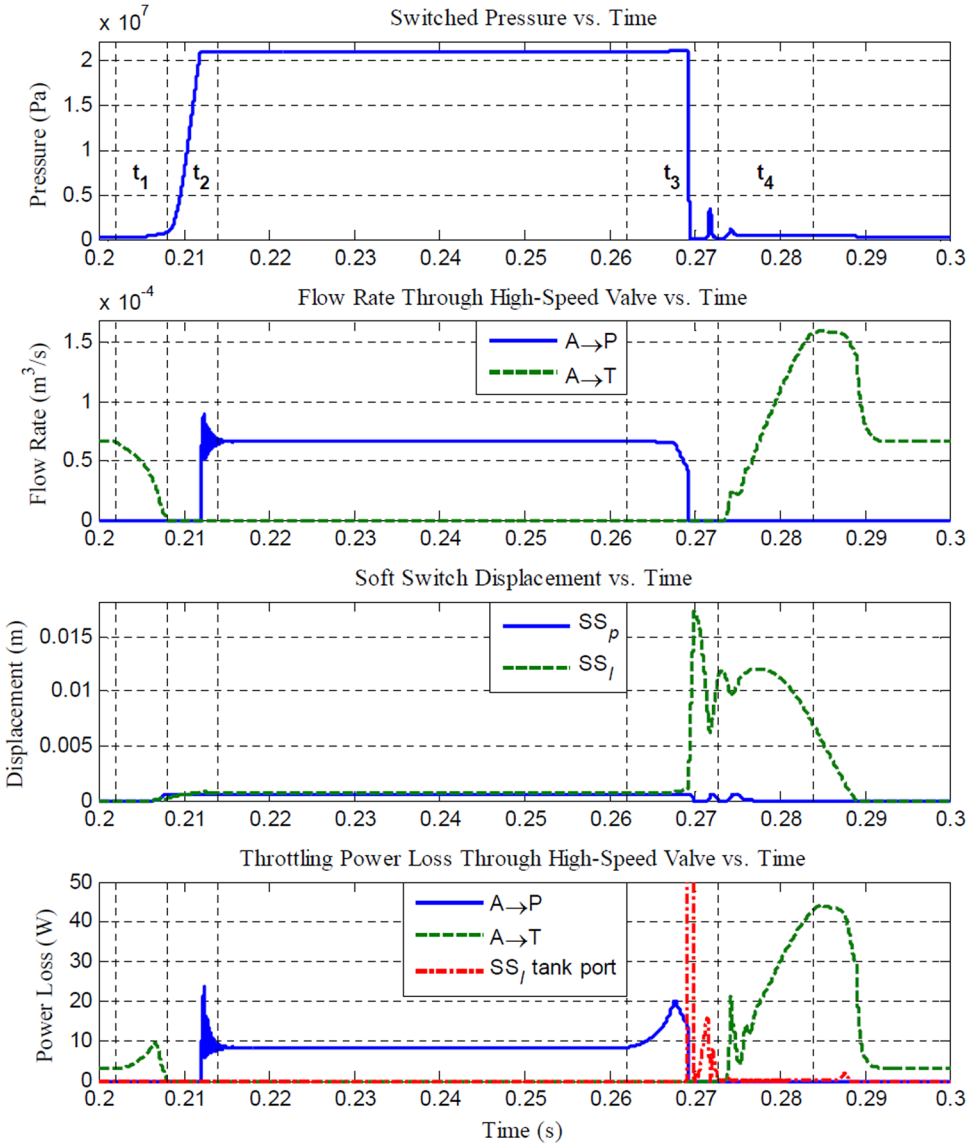

Fig. 8 Simulation results for the soft switch circuit using the diaphragm, Fig. 3(*a*), at 60% duty cycle

The large amplitude throttling loss across the locking soft switch tank port during the discharge of the diaphragm is addressed by eliminating the diaphragm and connecting the back side of the locking soft switch to the high pressure rail through a check valve, Fig. 3(*b*). As seen in Fig. 9, the switched volume pressure profile and the throttling power loss across the pressure and tank valves for this revised soft switch circuit is very similar to the previous circuit. The largest difference is the shorter duration is the spike in power loss across the locking soft switch tank port at the start of event t_3. This power loss is now only the result of decompressing the fluid on the back side of the switched volume and not emptying the diaphragm. During one second of operation at 60% duty cycle, the pump supplies 869.3 J of energy and the energy to the pressure rail is 835.6 J, for an energy loss fraction of 0.039.

The presence of a check valve in parallel with the pressure valve in the previous two soft switch circuits makes it challenging to reverse the flow and operate the pump as a motor. This issue can be addressed by removing the parallel check valve, which also eliminates the need for the passive soft switch, forming the circuit shown in Fig. 3(*c*). In the simulation results for this circuit, Fig. 10 similar behavior to the previous soft switch circuit is seen during events t_3 and t_4, while reverse flow through the pressure valve is observed during event t_2, as previously observed in the relief and check valve cases. The reason the power loss due to throttling through the pressure valve is greater than the relief valve case is that the volume on the back side of the soft switch adds to the effective compressible volume, resulting in more reverse flow through the pressure valve. For this circuit at 60% duty cycle, the energy delivered by the pump in one second is 877.7 J and the energy reaching the pressure rail is 827.9 J, for an energy loss fraction of 0.057.

To better compare the two control circuits and the three soft switching circuits, the energy loss during one second of operation as a function of the duty cycle is plotted in Fig. 11. The energy loss is defined as the difference between the energy supplied by the pump and the energy delivered to the high pressure rail. Recall that the three soft switch circuits were optimized for 75% duty cycle. The sharp increase in energy loss in these circuits above 75% duty cycle is due to the soft switches not emptying between cycles, preventing locking of the active soft switch.

Based on the quantity of fluid absorbed in the soft switches and whether a portion of the absorbed fluid is sent to the pressure

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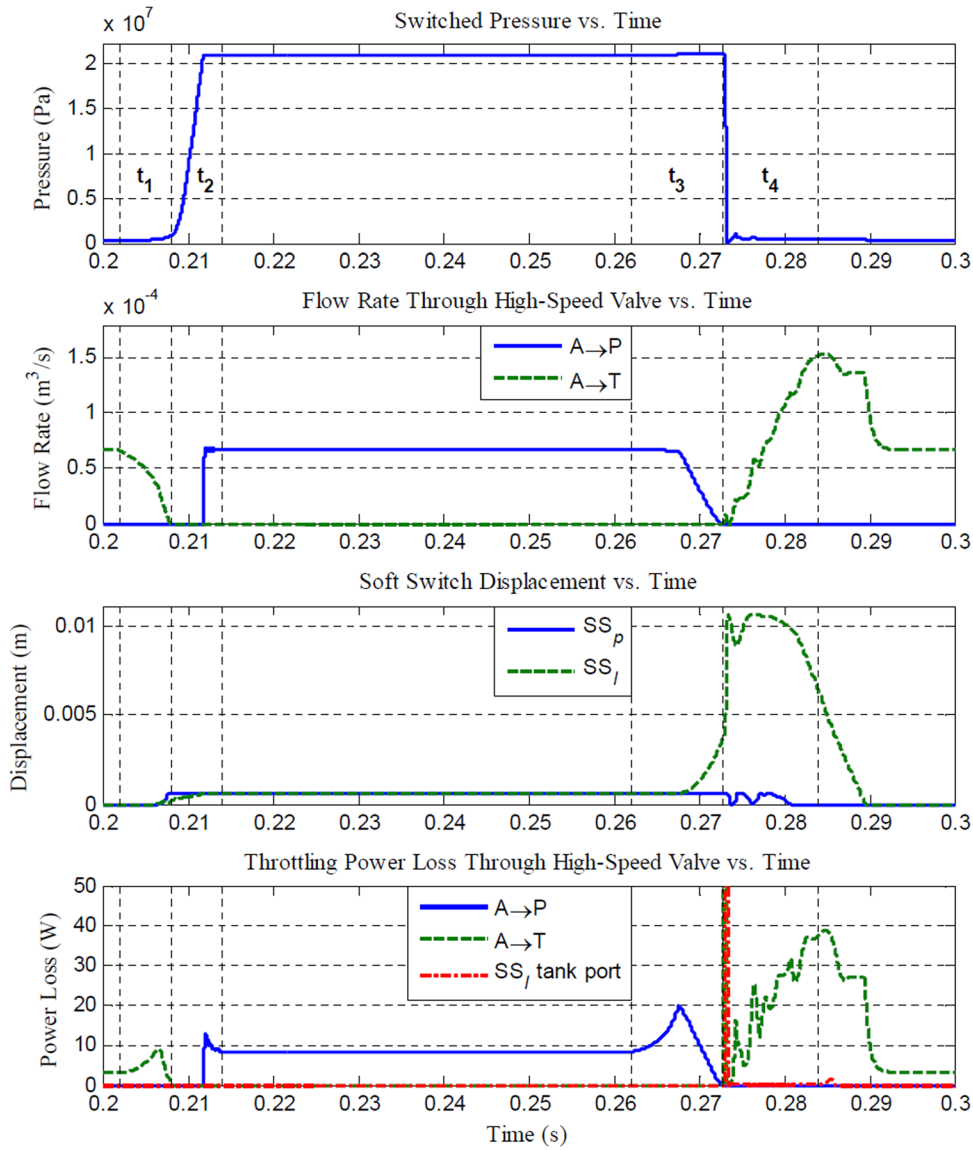

Fig. 9 Simulation results for the soft switch circuit illustrated in Fig. 3(*b*) at 60% duty cycle

branch, the energy delivered to the pressure branch at a specific duty cycle varies between circuits. To better understand the energy loss in a dimensionless form, the fraction of energy loss to the energy delivered versus the fraction of energy delivered to the energy delivered at 100% duty cycle is presented in Fig. 12. Note that at lower fractions of energy delivered, the energy loss fraction increases toward 1, yet the figure axes focus on the region of typical operation to improve the resolution.

5 Discussion

The simulation results revealed significant differences in the energy losses between the five virtually variable displacement pump circuits particularly, the transitional throttling losses. The majority of the energy losses associated with the pressure port transitions, t_2 and t_3, can be addressed with a check valve in parallel with the pressure valve, while the throttling across the tank ports requires soft switching.

Due to the parameters of the high-speed valve selected for this work, the energy losses during closing of the tank port, event t_1, are quite small in the control circuit using the relief valve, Fig. 6. There are three factors related to the parameters of the off-the-shelf high-speed valve that result in this small loss. As seen in Fig. 5, the valve has a shorter transition time for the first two transitions than the last two and has a larger orifice area for the connection between the pump and tank. Furthermore, the valve is slightly oversized for the pump flow rate. The impact of these valve characteristics is less need for the passive soft switch, illustrated in the optimization results by the short maximum travel of the passive soft switch piston in comparison to the locking soft switch piston, Table 1. The reader should recognize that this is not the general case as in other switch-mode circuits the first transition event can be more significant.

To further illustrate the influence of system operating conditions on the optimized soft switch geometry, a parameter optimization is again performed on the three soft switch circuits at a pump flow rate of 8 l/min, which is double the flow rate of the above results. When comparing the optimized parameters for higher flow rate, Table 2, with the base flow rate, Table 1, the travel of the soft switch pistons, particularly the passive piston is multiple times larger due to the relative pump flow rate to high-speed valve sizing. Further note that the required spring force increases significantly at the higher flow rate condition, despite operating at the same pressure. This is a result of the need to pass a higher pump flow and larger soft switch volume through the tank valve during a limited amount of time between cycles. It is

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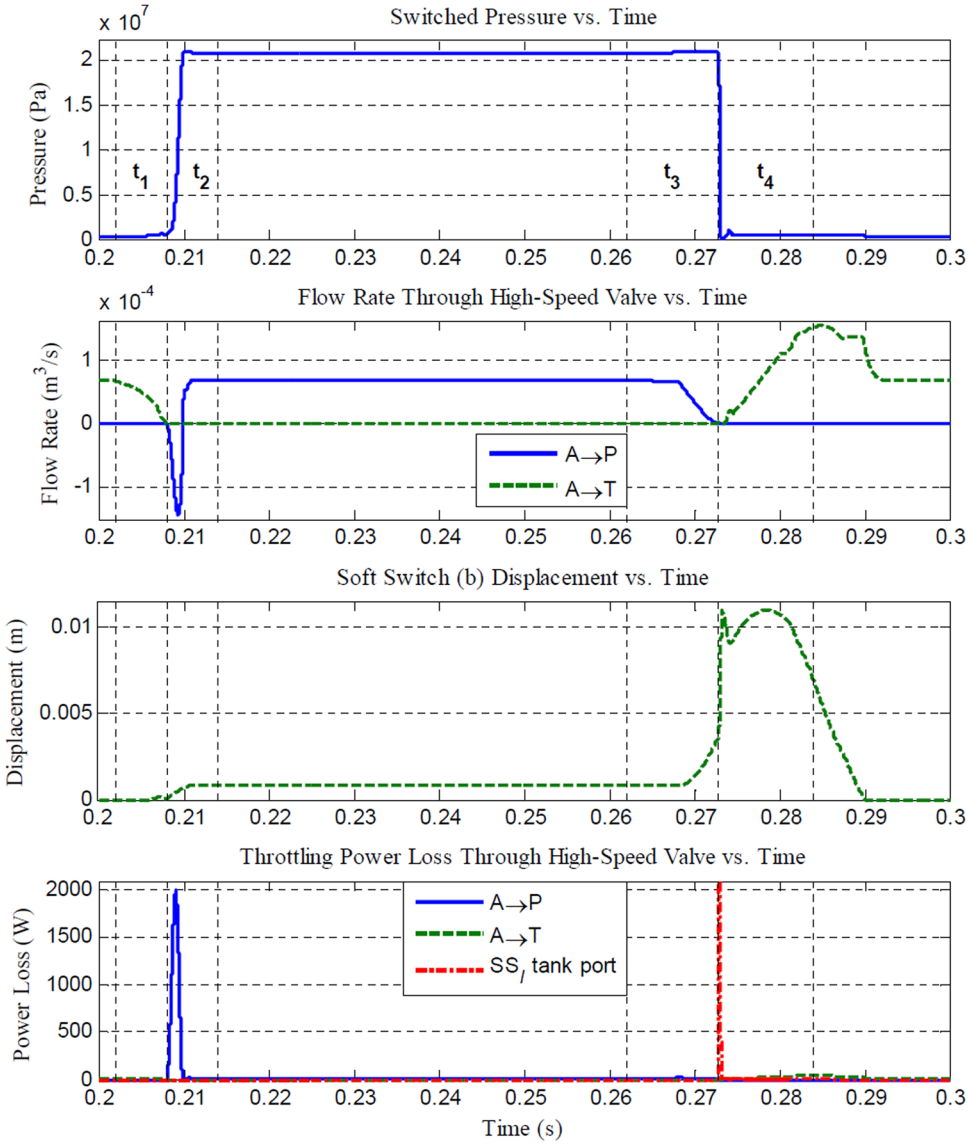

Fig. 10 Simulation results for the soft switch circuit illustrated in Fig. 3(*c*) at 60% duty cycle

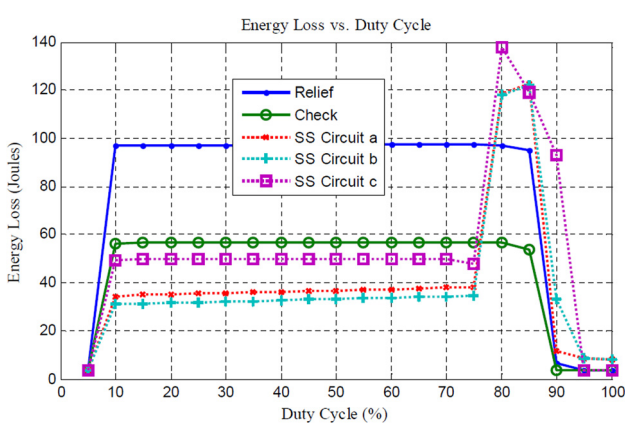

Fig. 11 Energy loss during one second of operation for the five circuits as a function of duty cycle

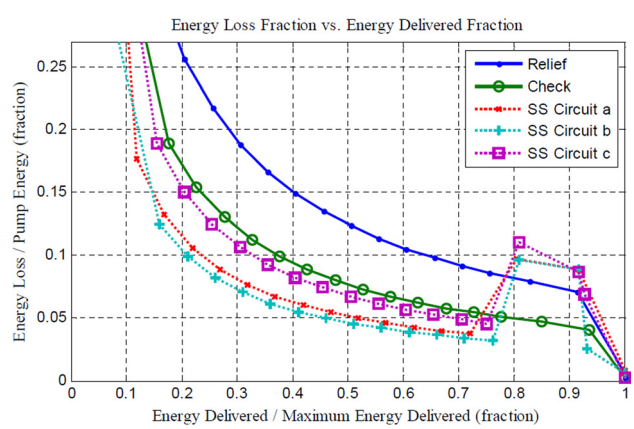

Fig. 12 Fraction of energy loss versus the fraction of energy delivered to the high pressure rail for the five circuits

interesting to note that for both flow rates and the different soft switch circuits, the optimal spring force for the passive and locking soft switch pistons was always the same, and thus only one value is reported.

Beyond the influence of the valve parameters used in this simulation, the required volume of the passive soft switch is less than the locking soft switch due to the compressibility of the fluid in the switched volume. In the 4 l/min case, the volume displaced by

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Table 2 Optimized soft switch parameters for a pump flow rate of 8 l/min, with all other factors the same as the previous optimization

Parameter	Symbol	Units	Value		
			Soft switch circuit a	Soft switch circuit b	Soft switch circuit c
Piston radius	$r_{ss,x}$	Mm	4.5	4.7	4.9
Max travel of passive piston	$x_{ss,a,\max}$	mm	3	2	0
Max travel of locking piston	$x_{ss,b,\max}$	mm	27.5	22.5	17.5
Spring preload force	$F_{\text{preload},x}$	N	64	66.7	76.7
Travel to open tank port on locking piston	$x_{\text{crack},sb}$	mm	1.5	4.3	4.4

the pump during the 12 ms time that transitions t_1 and t_2 occur is 0.8 cm^3, while the optimized volume of the passive soft switch in soft switch circuit b is 0.38 cm^3. During transitions t_3 and t_4, the displaced volume of the pump in 22 ms is 1.47 cm^3 and the optimized volume of the locking soft switch is 6.74 cm^3. During the first two transitions, the fluid in the switched volume is being compressed as the pressure is rising, decreasing the volume that needs to be stored in the soft switch. Conversely, the pressure in the switched volume is dropping during the second two transitions and the decompressing fluid significantly increases the volume requirement of the locking soft switch. Thus, the design of the soft switches is highly dependent on the fluid compressibility in the physical circuit as well as an accurate model of the fluid compressibility. As discussed in previous works, accurately modeling the effective bulk modulus in systems with rapid pressure transients is quite challenging [7,11].

In all of the circuits, the energy loss per second of operation is nearly constant between duty cycles of 10 and 75%, as seen in Fig. 11. Due to the soft switch circuit parameters being optimized for 75% duty cycle, the soft switches do not fully discharge at higher duty cycles, preventing the active soft switch from relocking between cycles. When the soft switch does not relock, the volume of the soft switches simply acts to increase the switched volume, increasing compressibility energy loss without the benefits of the flow absorption during throttling. This creates a sharp rise in energy loss at these higher duty cycles. A duty cycle of 75% was selected for optimization due to the valve timing; at this duty cycle, the tank valve is only closed for 3 ms out of the 100 ms cycle time. If the soft switches were optimized for a higher duty cycle, the return springs would need to be quite stiff to empty the soft switch chambers in the very short time connected to tank. The energy losses are low for all circuits at extremely high and low duty cycles because the valve does not make full transitions to tank or pressure, respectively.

While all three soft switch circuits lower the transitional throttling energy loss in comparison to the control circuits, there are significant differences between them. Soft switch circuits a and b address the throttling loss during all four valve transitions, while circuit c manages the throttling losses during the closing of the pressure valve and the opening of the tank valve, but suffers significant losses due to reverse flow through the pressure valve as it is opening. The energy loss results, Figs. 11 and 12, reveal this difference in that soft switch circuit c performs similar to the check valve circuit, while the other two soft switch circuits have significantly lower losses. Soft switch circuit b outperforms circuit a because the high pressure flow used to unlock the active soft switch is passed to the load through a check valve instead of being stored in the diaphragm and later throttled across the port to tank. Circuit b reduces the energy loss by an average of 66.1% across the range of functional duty cycles in comparison to the relief valve circuit.

The simulation presented in this work models the pressure dependent fluid compressibility in the switched volume and back side of the active soft switch, dynamics of soft switch piston due to pressure, spring, and frictional forces, and flow through the various valves with time varying orifice areas. The model makes a number of simplifying assumptions including neglecting the fluid inertia, assuming the outlet pressure is constant due to the accumulator, assuming the inlet flowrate is constant due to the inertia of the pump, assuming steady flow by using Bernoulli's equation to model the pressure-flow constituitive relationship of the valves, and assuming the check valves open instantaneously. Because the behavior of the check valves is important to the dynamics of the soft switches, the instant opening assumption should be examined in future work.

An additional energy loss incurred with the soft switches that has not been included in the above model is leakage past the soft switch pistons. If either soft switch piston experienced a pressure differential between the two sides of the piston equal to the difference between the accumulator pressure and tank pressure, the energy loss due to leakage, assuming laminar plate flow, is 22.2 J during one second of operation. This energy loss approaches the magnitude of the combination of the other energy losses. However, during operation, the active soft switch piston is only exposed to a significant pressure differential for a sub-millisecond amount of time as the soft switch port to tank is opening. The passive soft switch is only exposed to a large pressure differential when the piston has reached the end of the travel. At this location, the piston can be sealed against the seat, preventing leakage. For these reasons, the leakage past the soft switch pistons during actual operation is minimal, and thus neglected in the above model.

6 Conclusion

In this paper, a novel locking soft switch design for use in switch-mode circuit, which uses a pressure signal for release, is presented. Three unique circuit configurations using the locking soft switch in a virtually variable displacement pump system were simulated, optimized, and compared with two conventional circuits using a relief valve and check valve in parallel with the pressure valve to manage pressure spikes. The average energy loss reduction, using the relief valve circuit as a reference, across a duty cycle range from 15% to 75% was 41.7% for the check valve circuit, 62.4% for soft switch circuit a, 66.1% for soft switch circuit b, and 49.0% for soft switch circuit c. Soft switch circuit b outperformed the other circuits by reducing the throttling loss of all four valve transitions and passing the hydraulic energy used to unlock the soft switch to the load.

While the soft switch parameters for the three circuit configurations were optimized for a specific set of operating conditions, the solutions are flexible to a certain degree. The soft switches perform properly at lower duty cycles, yet at higher duty cycles the system is not connected to the tank port for sufficient time to empty the soft switch chambers. Similarly, the soft switch circuits operate properly at pressures lower than the design pressure, yet at higher pressures, the return spring does not provide sufficient force to empty the chambers between cycles. When the soft switch chambers do not empty, the lock does not re-engage and the soft switch simply acts as additional compressible volume.

Future work will include detailed design of a soft switch and experimental studies. As the operation of the soft switch was found to be sensitive to the geometric parameters, it is expected that experimental tuning will be required to achieve proper operation. For example, manufacturing tolerances make it challenging to create the soft switch piston to sleeve clearance specified in the model. As leakage is a cubic function of this clearance, small variations will create results that diverge from modeled behavior.

The locking soft switch design presented in this paper significantly reduces energy losses in switch-mode hydraulic circuits, being applicable to pump, motor, transformer, and actuator

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applications. Because the soft switch prevents throttling across the transitioning valve, the transition speed requirements of the valve become less important. As valve transition time is typically a function of the energy consumed by the valve, this is an important contribution to the field of switch-mode hydraulics.

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